In terms of this atomic mass unit, the mass of the neutron = 1.008665 u, the mass of the proton =1.007276 u, the mass of the hydrogen atom = 1.007825 u and the mass of ${}^{12}_{6}C = 12u$. Note that usually mass specified is the mass of the atom, not mass of the nucleus. To get the nuclear mass of particular nucleus, the mass of electrons has to be subtracted from the corresponding atomic mass. Experimentally the atomic mass is determined by the instrument called Bainbridge mass spectrometer. If we determine the atomic mass of the element without considering the effect of its isotopes, we get the mass averaged over different isotopes weighted by their abundances.

EXAMPLE 9.6

Calculate the average atomic mass of chlorine if no distinction is made between its different isotopes?

Solution

The element chlorine is a mixture of 75.77% of ${}^{35}_{17}Cl$ and 24.23% of ${}^{37}_{17}Cl$. So the average atomic mass will be

$$\frac{75.77}{100} \times 34.96885u + \frac{24.23}{100} \times 36.96593u$$
$$= 35.453u$$

In fact, the chemist uses the average atomic mass or simply called chemical atomic weight (35.453 u for chlorine) of an element. So it must be remembered that the atomic mass which is mentioned in the periodic table is basically averaged atomic mass.

9.4.4 Size and density of the nucleus

The alpha particle scattering experiment and many other measurements using different methods have been carried out on the nuclei of various atoms. The nuclei of atoms are found to be approximately spherical in shape. It is experimentally found that radius of nuclei for $Z > 10$, satisfies the following empirical formula

$$R = R_0 A^{\frac{1}{3}} \qquad (9.19)$$

Here A is the mass number of the nucleus and the constant R_0 = 1.2 F, where 1 F = 1×10^{-15} m.The unit fermi (F) is named after Enrico Fermi.

EXAMPLE 9.7

Calculate the radius of ${}^{197}_{79}Au$ nucleus.

Solution

According to the equation (9.19),

$$R = 1.2 \times 10^{-15} \times (197)^{\frac{1}{3}} = 6.97 \times 10^{-15} m$$

Or R = 6.97 F

EXAMPLE 9.8

Calculate the density of the nucleus with mass number A.

Solution

From equation (9.19), the radius of the nucleus, $R = R_0 A^{\frac{1}{3}}$. Then the volume of the nucleus

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^3 A$$

By ignoring the mass difference between the proton and neutron, the total mass of the nucleus having mass number A is equal to $A.m$ where m is mass of the proton and is equal to 1.6726 x 10^{-27} kg.

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